

Tagging the decays of the Z^0 boson into b quark pairs with a neural network classifier

C. Bortolotto, A. De Angelis and L. Lanceri

Istituto di Fisica dell'Università di Udine and INFN Trieste, Trieste, Italy

Received 17 December 1990 and in revised form 8 March 1991

By using a neural network classifier, we are able to identify simulated decays of the Z^0 boson into $b\bar{b}$ pairs with $\approx 85\%$ efficiency, and with a fraction of misidentified light quark decays of $\approx 45\%$. The simulation takes into account the experimental inefficiencies and resolutions typical of a LEP detector, and makes use of the reconstruction of charged tracks only.

1. Introduction

This article presents a study on the identification of the decays of the Z^0 into $b\bar{b}$ pairs, using a neural network identifier. This technique has been successfully used for distinguishing quark and gluon jets [1]. Preliminary works aiming at the separation of samples enriched in $b\bar{b}$ events have been reported [2], although reaching low efficiencies and assuming a perfect detector.

The tagging of $b\bar{b}$ events has a primary importance, both for preparing as pure as possible subsamples of events for b studies, and for determining the partial width of the Z^0 into $b\bar{b}$ pairs. This last quantity is especially sensitive to the top mass, and its precise determination is one of the major aims of LEP in the next years [3].

We will show that techniques based on neural networks can complement the results reachable by linear multidimensional analysis [4].

2. The simulation

2.1. Simulation of the final state

The simulation is based on events generated by means of the JETSET 7.2 Parton Shower Monte Carlo (JETSET PS [5]), that has proven, after one year of activity of LEP, to reproduce the shape of the hadronic decays of the Z^0 well [6].

Modifications were made with respect to the default in order to obtain a better agreement with the experi-

mental data on b fragmentation. Recent results from L3 [8], ALEPH [9] and OPAL [10] allow to constrain the fragmentation of b at $\sqrt{s} = 91.2$ GeV.

The average energy fraction $\langle x_B \rangle = \langle E_B \rangle / E_{\text{beam}}$ of bottom hadrons, from the average of the three determinations, comes out to be

$$\langle x_B \rangle = 0.703^{+0.018}_{-0.013}. \quad (1)$$

JETSET PS, with the default symmetric fragmentation functions (defined as $D(z) = (1/z)(1 - z)^a \exp(-bm_T^2/z)$, where m_T is the hadron transverse mass, $z = (E + P_L)_{\text{had}} / (E + P)_{\text{quark}}$ and $a = 0.5$, $b = 0.9 \text{ GeV}^{-2}$), predicts a value $\langle x_B \rangle = 0.755$, that is ≈ 3 standard deviations higher than current measurements.

Switching to Peterson fragmentation functions for heavy quarks (defined as

$$D_Q^H(z) = N \left(\frac{1}{z} \right) \left[1 - \left(\frac{1}{z} \right) - \epsilon_Q / (1 - z) \right]^{-2},$$

where the normalization N is fixed by summing over all hadrons containing the heavy quark Q [7]), the fragmentation is described by a parameter ϵ_b . The relationship between ϵ_b and the average energy fraction is not trivial, being the fragmentation process mediated by the parton showering (the variable z in the fragmentation function cannot be directly accessed experimentally). It can however be computed by assuming a model for the parton showering process. By assuming JETSET PS with default parameters, the value of ϵ_b , corresponding to that of $\langle x_B \rangle$ in eq. (1), is

$$\epsilon_b = (3.2 \pm 1.6) \times 10^{-3}. \quad (2)$$

The problem of the determination of the c fragmentation function (that could be relevant for estimating

Table 1

Effects of experimental resolution and acceptance in the Monte Carlo program (values averaged over the performance of the 4 LEP detectors).

Charged tracks	
minimum observable momentum	100 MeV/ c
minimum polar angle	30°
momentum resolution	$\sigma(p_T)/p_T = 0.002 p_T$
Muon identification	
momentum threshold	2 GeV/ c
efficiency	85%
hadron misidentification probability	1%
Vertex detector	
impact resolution (xy plane)	$\sigma_{xy} \approx 200 \mu\text{m}$

the backgrounds) is, at present, discussed only in ref. [9]. The ALEPH Collaboration measured:

$$\langle x_C \rangle = 0.52^{+0.11}_{-0.12} \quad (3)$$

and obtained (for $c\bar{c}$ events)

$$\epsilon_c = (24^{+47}_{-22}) \times 10^{-3}. \quad (4)$$

2.2. Simulation of the detector

The response of the detector has been simulated according to smearing parameters summarized in table 1. These parameters come from the study of an “average” LEP detector in ref. [11]. We assume – pessimistically – that only charged particles are reconstructed, and we do not make use of kaon identification.

After one year of activity of LEP, the performance of table 1 is fully verified for what is related to momentum reconstruction for charged tracks, slightly optimistic for what is related to muon identification.

3. Choice of the variables

The variables used for the discrimination of $b\bar{b}$ events are:

1) The number of jets. Jets were defined by JADE/E0 algorithm [12]. In this algorithm, for all pairs of particles k, l of an event, the scaled invariant mass $y_{kl} = M_{kl}^2/E_{\text{vis}}^2$ is computed (E_{vis} is the seen energy in the event) and the two particles with the smallest y_{kl} value are replaced by a “cluster” of 4-momentum ($p_k + p_l$). This procedure is iterated until all y_{kl} exceed a threshold value y_{cut} and the resulting number of clusters is called the jet-multiplicity of the event [12]. We used $y_{\text{cut}} = 0.08$.

2) The boosted sphericity product [13,14]. We define this variable as the product of the sphericities of the two

most energetic jets, after a boost $\beta = 0.96$ along their axis.

3) The transverse mass of the event [15]. This variable is defined as $2E_{\text{beam}} p_{\text{t}}^{\text{out}}/E_{\text{vis}}$, where $p_{\text{t}}^{\text{out}}$ is the sum of the absolute values of the momentum components perpendicular to the aplanarity plane.

4) The momentum component perpendicular to the jet axis of the muon for which this quantity is largest.

5) The sum of the absolute values of the impact parameters in the plane perpendicular to the beam (xy plane), each one divided by its expected error. The microvertex detector is simulated according to the method outlined in ref. [16]. As a result, the rms of the distribution of the impact parameters in the plane perpendicular to the beam axis is $\approx 200 \mu\text{m}$.

6) The second factorial moment [17] of the rapidity distribution with respect to the sphericity axis, corresponding to a rapidity interval $\delta y = 0.1$. This variable is sensitive to the possible presence of peaks in the rapidity distribution of $b\bar{b}$ events [18].

7) The momentum p , the polar angle θ and the azimuth ϕ (with respect to the sphericity axis) of the four most energetic charged tracks in the most energetic jet.

The variables mentioned above have been normalized in such a way that they range from 0 to 1.

Some of the variables have already been used in literature in problems related to the identification of $b\bar{b}$ events [4,13–15]. The use of a microvertex detector is pessimistic, in the sense that it has not been hypothesized that such a device is used to reconstruct secondary vertices, but only to measure impact parameters. Reconstructing secondary vertices would of course allow to separate an almost pure sample of unresolved $b\bar{b}$ and $c\bar{c}$ events.

4. The neural network

To classify the simulated hadronic decays of Z^0 as “ $b\bar{b}$ events” or “non- $b\bar{b}$ events” a neural network approach [19] was adopted.

We chose a feed-forward layered network [20] with 18 nodes in the input layer, associated with the input variables x_i , defining the pattern space P ; 10 nodes in the hidden layer; and one output node, associated with the output value Θ , belonging to the feature space F . The number of neurons in the hidden layer, used by the network for its internal representation, was chosen as an intermediate one between the number of input and output neurons. It was checked a posteriori that the performance of the network is stable around the number of 10 hidden neurons. In particular, we did not notice an appreciable loss in performance by decreasing the number of hidden units by three. When increasing this number by three, the network was able to adjust

part of the weights in such a way to remove the superfluous connections.

Each neuron (node) performs a weighted sum of the output values from all the nodes of the previous layer; the node output is then computed via a sigmoid function

$$g_T(x) = \frac{1}{1 + e^{-x/T}} \quad (5)$$

at a “temperature” T ^{#1}. The output o_i of the i th neuron of a layer (starting from the second) is then

$$o_i = g_T\left(\sum_j \omega_{ij} o_j\right), \quad (6)$$

where the sum is made over the nodes of the previous layer.

All input variables are normalized so that they can vary from 0 to 1. One can therefore compare their relative importance by comparing the corresponding weights.

4.1. The network training

The network training [20] procedure fixes the values of the weights associated with the node interconnections. The aim is to realize a mapping of the input pattern space ($x_i \in P$) to the feature space ($\Theta \in F$), such that a good separation of “ $b\bar{b}$ ” from “non- $b\bar{b}$ ” events is obtained by a simple cut in Θ .

In the “back-propagation” learning algorithm the output feature values for the “training” input patterns are computed and compared with the corresponding desired “target” values. A least mean square error function E is computed to quantify the difference between the obtained output Θ and the desired target t :

$$E = \frac{1}{2} \sum_{\text{patterns}} (t - \Theta)^2. \quad (7)$$

This function is minimized by changing (“updating”) the weights by an amount computed from the error function by the gradient descent method [20]. The process is controlled by the “learning strength parameter” η and the “momentum” α [20]: each updating step in the space of weights, computed by gradient descent, is multiplied by η and added to the previous step, multiplied by α . To smooth out fluctuations, weights are updated using the cumulative error from a number of input training patterns (10 in our case). A sequence of

10 patterns will be simply called an “update” in the following.

During the training procedure, we can change the learning speed by changing η and α after a set of input patterns (10 000 in our case), that we call “learning epoch”. Decreasing η slows down learning, but has the positive effect of dumping fluctuations. Increasing α has a similar effect, increasing the inertia of the network against large changes in the weights. Our final choices for these parameters are discussed in section 5.

For the training of the system, we generated by Monte Carlo, according to the prescriptions described in section 2, a set of 100 000 Z^0 decays into $b\bar{b}$ pairs and an equal total number of $u\bar{u}$, $d\bar{d}$, $c\bar{c}$, $s\bar{s}$ events (in the proportions predicted by the Standard Model). The training was made with an equal number of signal and background events, because we verified that this improves the performance of the network.

We defined two symmetric target values t_1 for “non- $b\bar{b}$ ” and $t_2 = 1 - t_1$ for “ $b\bar{b}$ ” input patterns. We tried different choices of t_1 , between 0 and 0.5, in steps of 0.05. The performances of the network were evaluated after learning with each of these target values. The method and results are discussed in section 4.2 and 5.

4.2. Testing the network

At each step of the learning procedure, an indication on the performance of the network can be inferred from the error function (7). A more reliable evaluation is obtained by testing the response of the network on a set of input patterns independent of the training set. Two sets of test patterns were thus generated by the Monte Carlo programs described in section 2, for “ $b\bar{b}$ ” events, and “non- $b\bar{b}$ ” events respectively. In this case, the populations of the two samples were not equal as in the training phase, but the fraction of $b\bar{b}$ events was chosen equal to 0.217, as predicted by the Standard Model.

The simplest approach is to classify as “ $b\bar{b}$ candidates” all the events for which the network output value in the feature space is $\Theta > \Theta_{\text{cut}}$ and as “non- $b\bar{b}$ candidates” all those for which $\Theta < \Theta_{\text{cut}}$. A reasonable guess for the choice of Θ_{cut} in our case is $(t_1 + t_2)/2 = 0.5$. We systematically studied the network efficiencies for $0 < \Theta_{\text{cut}} < 1$.

The “signal efficiency” ϵ_S was measured as the ratio:

$$\epsilon_S = N_S^a / N_S, \quad (8)$$

where N_S^a is the number of patterns accepted by the ($\Theta > \Theta_{\text{cut}}$) criterion in a sample of N_S input patterns of “ $b\bar{b}$ ” type. The “purity” p was defined as:

$$p = N_S^a / [N_S^a + N_B^a], \quad (9)$$

where N_B^a is the number of patterns, accepted by the same criterion, from a sample of N_B background pat-

^{#1} The sigmoid function squeezes the node output between 0 and 1; the final output becomes more peaked at 0 or 1 as the temperature decreases, becoming exactly 0 or 1 in the limiting case in which the temperature is zero.

terns. The purity can be interpreted as the fraction of genuine $b\bar{b}$ events to be found in a mixed sample, selected by the criterion $\Theta > \Theta_{\text{cut}}$, if the input events are a mixture of signal and background in the Standard Model proportion $N_S/[N_S + N_B] = 0.217$.

5. Analysis and results

The study of the network led to the choice of the values of the network parameters T , η and α summarized in table 2.

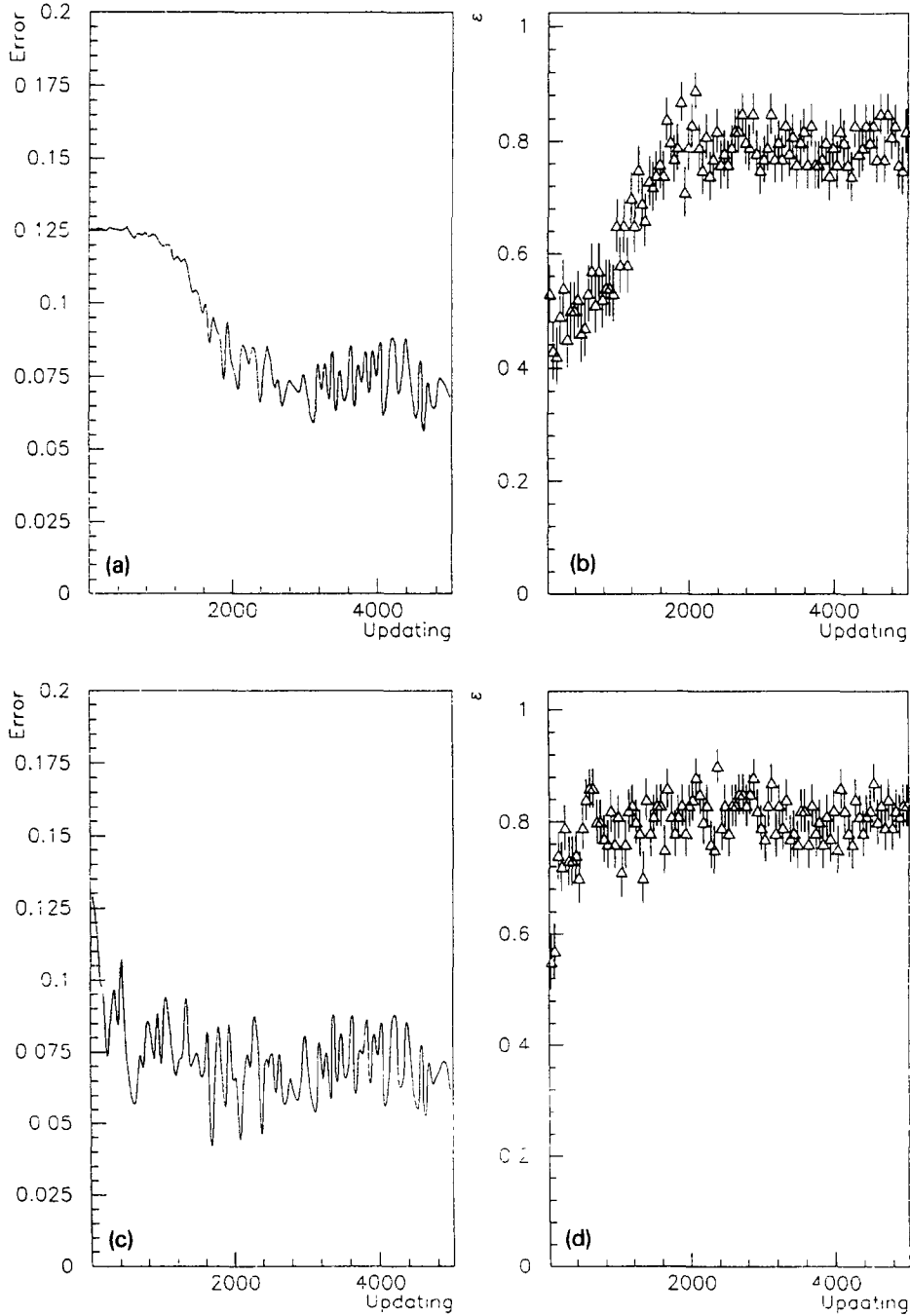


Fig. 1. For $t_1 = 0.05$, $t_2 = 0.95$, and “fixed” learning mode (see text): (a) Error function in the first 5000 updatings (50000 input patterns). (b) Fraction ϵ of correct identifications as a function of the updating. (c) and (d) same as (a) and (b), for the “variable” learning mode.

We present results on the learning performance of the network in two different learning modes. In the first, all three parameters are kept fixed at the values indicated in table 2 under the “fixed mode” heading. In the second, η is decreased and α is increased at each new learning epoch, as discussed in section 4.1, from the

initial to the final values reported in table 2, under the “variable mode” heading. In both modes, the target values are $t_1 = 0.05$ and $t_2 = 0.95$.

The learning performance of the network is shown in figs. 1a and 1b for the “fixed” learning mode, and in figs. 1c and 1d for the “variable” learning mode. Fig. 1a

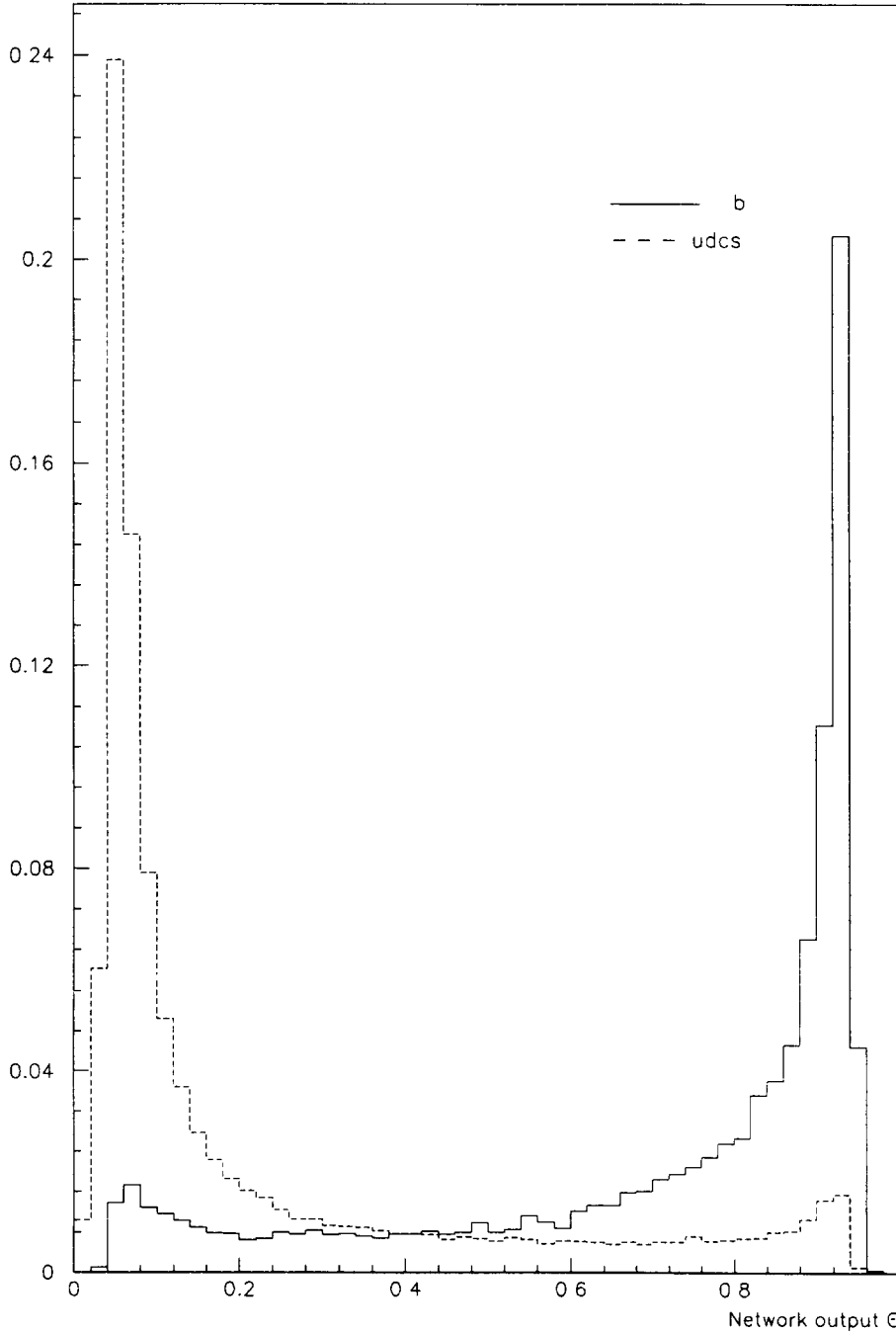


Fig. 2. Probability density function for the network output after full training, for $b\bar{b}$ (solid line) and non- $b\bar{b}$ (dashed line) events. $t_1 = 0.05$, $t_2 = 0.95$.

Table 2

Two different choices of the detector parameters (see text). In the second column, η and α vary during the learning phase.

Parameter	"Fixed" mode	"Variable" mode
T	1	1
η	0.01	0.5–0.01
α	0.5	0.5–0.8

shows the evolution of the error function in the first 5000 updatings (50 000 input patterns). Correspondingly, in fig. 1b the fraction ϵ of correct identifications with $\Theta_{\text{cut}} = 0.5$ (irrespective of the appartenance to the signal or background sample) is plotted as a function of the updating. These plots show that the network learning reaches an efficiency plateau at about 85% after $\approx 20\,000$ simulated events, with a corresponding drop

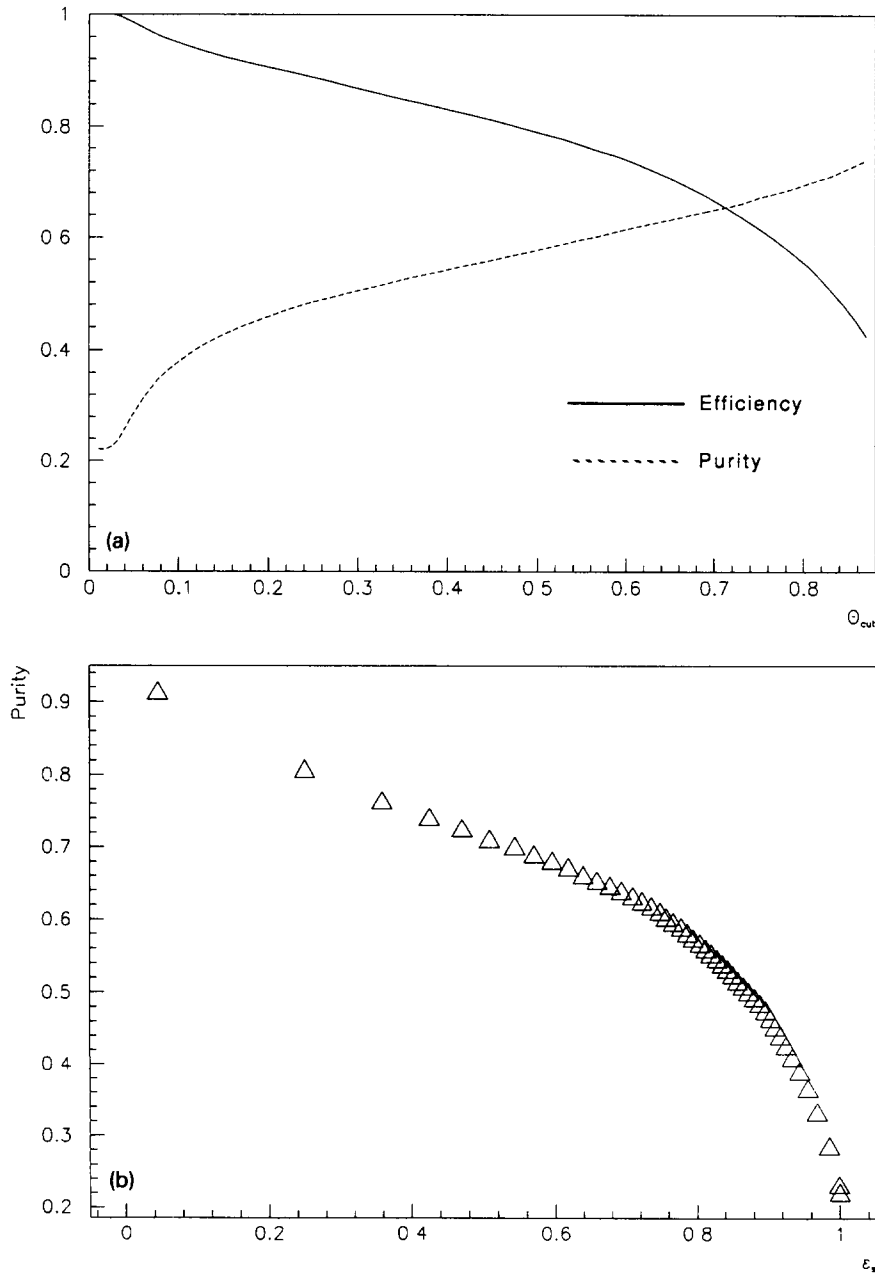


Fig. 3. For $t_1 = 0.05$, $t_2 = 0.95$, after full training: (a) Efficiency of signal identification (solid line) and purity of the selected sample (dashed line) as a function of Θ_{cut} . (b) Purity versus efficiency.

in the error function from the initial 0.125 to a plateau value ≈ 0.075 .

Figs. 1c and 1d illustrate the evolution of the error function and of the above defined efficiency in the “variable” learning mode. The final plateau values are almost the same as those of the “fixed” mode, but they are approached faster (after less than 10000 simulated events).

The probability distributions of the network output Θ after full training in “variable” mode are plotted in

fig. 2, displaying a good separation between $b\bar{b}$ (solid line) and non- $b\bar{b}$ (dashed line) events. The integrals of both distributions of the network outputs Θ for “ $b\bar{b}$ ” and “non- $b\bar{b}$ ” input patterns are normalized to 1.

The efficiency of signal identification and the purity of the selected sample as a function of Θ_{cut} are plotted in fig. 3a; in fig. 3b the purity is plotted vs the efficiency.

This computation was repeated systematically varying the target $t_1(t_2)$ in the training phase between 0 and

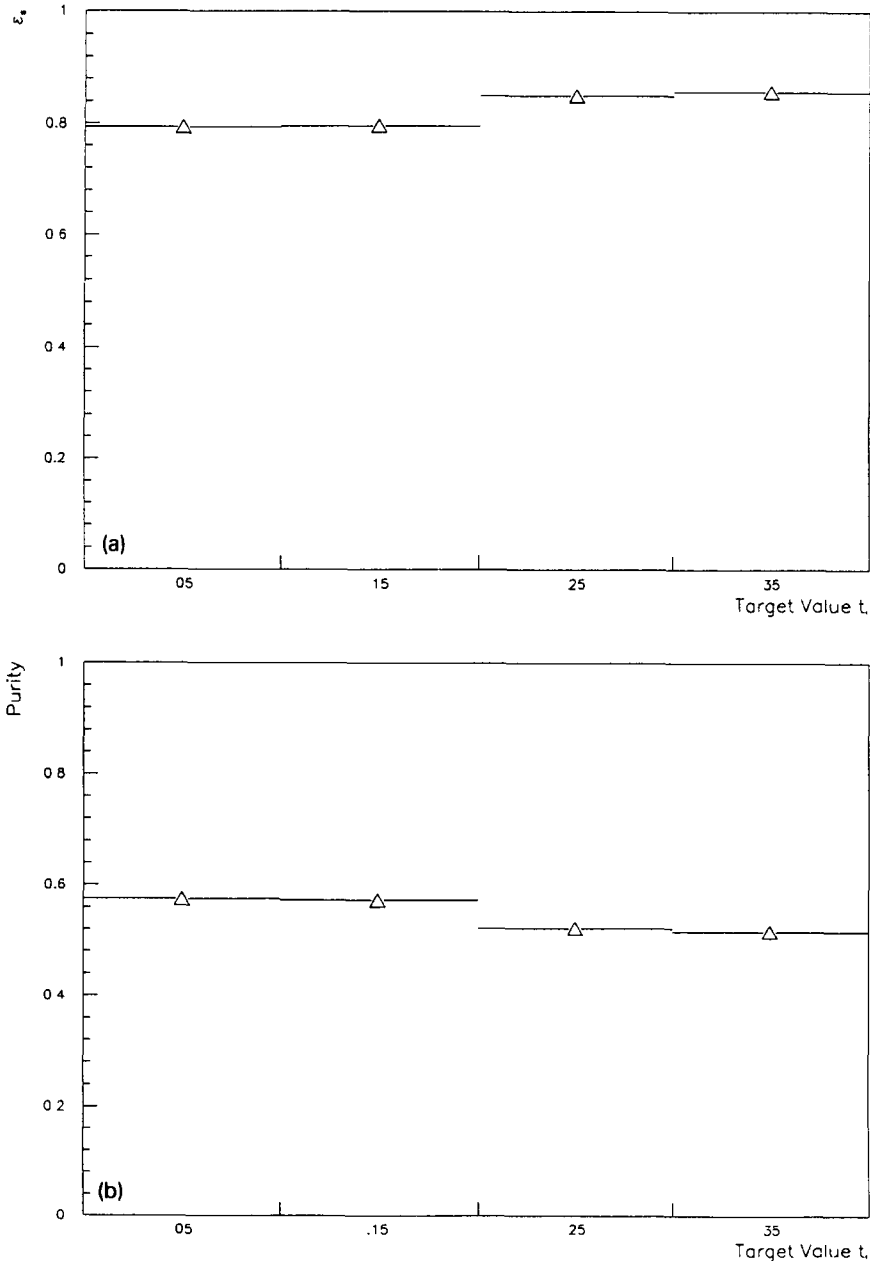


Fig. 4. Final purity (a) and efficiency (b) for a cut $\Theta_{\text{cut}} = 0.5$, as a function of the target t_1 in the training.

0.5 (1 and 0.5) in steps of 0.05. The network performance comes out to be almost independent of the target value for $t_1 < 0.4$ (fig. 4). In this region, we can achieve an efficiency for $b\bar{b}$ tagging in the order of $\approx 85\%$ with a purity of $\approx 55\%$, if $\Theta_{\text{cut}} = 0.5$.

Some indications on the relative importance of the input variables chosen by us is given by the weights obtained by the network after the learning procedure. We could not identify any input variable systematically associated with very small weights. There is only some indication that the node associated with the variable number 6 may be less important than the others. The reproducibility of the network performance was checked by repeated training sessions using different random initial values for weights and independent samples of training patterns.

6. Conclusions

We have shown that a neural network classifier can be used to tag the decays of the Z^0 boson into b quark pairs.

Our study is based on the generation of events by means of the JETSET 7.2 Parton Shower Monte Carlo and on the simulation of a realistic "average" LEP detector. We selected 18 variables as inputs to a layered feedforward network. After a detailed study of the network parameters, we were able to train it in a stable and reproducible way, to identify simulated Z^0 decays into $b\bar{b}$ pairs with $\approx 85\%$ efficiency, and a fraction of $\approx 45\%$ of misidentified light quarks. The purity of the sample can be substantially increased, of course at the expense of a lower tagging efficiency.

These results look very promising, and are competitive with and complementary to those obtained by multidimensional analysis methods.

Acknowledgements

We should like to thank O. Barring for introducing us to the technology of feedforward networks and C. Peterson for kindly giving explanations on some formulae in his papers on the subject.

References

- [1] L. Lönnblad et al., Phys. Rev. Lett. 65 (1990) 1321; and C. Peterson, Neural Networks and High Energy Physics, LU TP 90-6, Lund (1990).
- [2] L. Lönnblad et al. Nucl. Phys. B349 (1991) 675.
- [3] P. Mättig, A High Luminosity LEP as a B-Factory, CERN-EP/90-71 (1990).
- [4] C. de la Vaissiere and J. Palma Lopez, Multidimensional analysis: a tool for b tagging, DELPHI CERN 89-5 PHYS 5 (1989).
- [5] T. Sjöstrand, Comput. Phys. Commun. 27 (1982) 243, ibid. 28 (1983) 229; and T. Sjöstrand and M. Bengtsson, Comput. Phys. Commun. 43 (1987) 367.
- [6] M.Z. Akrawy et al. (OPAL collaboration), Z. Phys. C47 (1990) 505; and P. Aarnio et al. (DELPHI collaboration), Phys. Lett. B240 (1990) 271.
- [7] C. Peterson et al., Phys. Rev. D27 (1983) 105.
- [8] B. Adeva et al. (L3 collaboration), Phys. Lett. B241 (1990) 416.
- [9] D. Decamp et al. (ALEPH collaboration), Phys. Lett. B244 (1990) 551.
- [10] OPAL collaboration, contributed paper to the Singapore High Energy Physics Conf. 1990, presented by A. Jawahery.
- [11] Proc. ECFA Workshop on LEP200, eds. A. Böhm and W. Hoogland, CERN 87-08 (1987).
- [12] W. Bartel et al. (JADE collaboration), Z. Phys. C33 (1986) 23.
- [13] W. Braunschweig et al. (TASSO collaboration), Z. Phys. C44 (1989) 1.
- [14] P. Abreu et al. (DELPHI collaboration), CERN-PPE/90-118, Geneva (1990).
- [15] R. Marshall, Z. Phys. C26 (1984) 291.
- [16] R. Settles, in: Physics at LEP, eds. J. Ellis and R. Peccet, CERN 86-02 (1986) vol. 2, p. 214.
- [17] A. Bialas and R. Peschanski, Nucl. Phys. B273 (1986) 703.
- [18] A. Giovannini, R. Ugoccioni and L. van Hove, Proc. Int. Workshop on Correlations and Multiparticle Production (CAMP), Marburg 1990, eds. M. Plümer, S. Rana and R.M. Weiner (World Scientific, 1991) p. 219.
- [19] E. Domany, J. Stat. Phys. 51 (1988) 743.
- [20] D.E. Rumelhart, G.E. Hinton and R.J. Williams, Learning Internal Representations by Error Propagation, in: Parallel Distributed Processing: Explorations in the Microstructure of Cognition, vol. 1, eds. D.E. Rumelhart and J.L. McClelland, (MIT Press, 1986).